

BLAGOVESHENKA, Kazakhstan

WMO: 287660

Lat: 54.3589N

Lon: 66.9801E

Elev: 151

StdP: 99.52

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-30.3	-37.0	0.1	-33.6	-33.7	0.2	-30.2	12.4	-8.7	12.0	-8.0	1.1	230	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	31.0	18.8	29.0	18.4	27.2	17.7	20.7	27.4	19.8	26.1	18.9	25.0	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.5	23.1	17.6	12.8	22.3	16.7	12.1	21.6	60.3	27.4	57.3	26.1	54.4	25.1	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	9.9	8.3	-36.8	33.7	3.6	2.3	-39.3	35.4	-41.4	36.8	-43.4	38.1	-46.0	39.8		
(5)			-36.9	22.4	3.5	0.9	-39.4	23.0	-41.4	23.5	-43.4	24.0	-45.9	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-17.4	-15.7	-7.1	4.3	12.9	18.0	19.1	17.4	11.6	4.0	-5.9	-13.9	(6)
(7)		DBStd	14.39	7.29	7.27	5.77	6.04	4.91	4.25	3.51	4.09	4.60	4.77	7.23	7.54	(7)
(8)		HDD10.0	3759	849	719	529	187	28	2	0	1	33	192	477	742	(8)
(9)		HDD18.3	5979	1108	953	787	421	176	57	34	67	206	443	727	1000	(9)
(10)		CDD10.0	979	0	0	0	16	119	241	282	231	82	8	0	0	(10)
(11)		CDD18.3	157	0	0	0	0	8	46	57	39	6	0	0	0	(11)
(12)		CDH23.3	1692	0	0	0	4	163	519	534	392	78	0	0	0	(12)
(13)		CDH26.7	501	0	0	0	1	36	171	157	117	20	0	0	0	(13)
(14)	Wind	WSAvg	3.0	3.0	3.3	3.4	3.3	3.3	2.7	2.3	2.3	2.7	3.1	3.1	3.2	(14)
(15)	Precipitation	PrecAvg	362	17	13	18	23	28	46	63	47	32	29	25	21	(15)
(16)		PrecMax	524	32	33	41	65	67	151	128	134	65	61	50	48	(16)
(17)		PrecMin	202	1	1	1	2	3	8	16	3	4	3	2	4	(17)
(18)		PrecStd	77	9	8	11	15	15	31	33	31	16	15	12	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	1.2	8.1	23.5	30.1	32.9	33.1	32.4	29.2	20.3	8.8	1.2	(19)
(20)			MCWB	-0.8	-0.1	4.8	13.0	16.6	19.0	19.5	18.9	15.9	11.2	6.2	-0.1	(20)
(21)		2%	DB	-2.1	-0.7	3.2	19.4	27.2	30.9	30.2	29.9	25.5	16.9	5.8	-0.1	(21)
(22)			MCWB	-2.8	-1.5	1.5	11.4	15.3	18.7	19.8	18.6	14.5	10.0	4.5	-1.0	(22)
(23)		5%	DB	-3.8	-2.6	2.0	16.7	25.0	28.6	28.2	27.5	22.2	14.1	4.1	-1.9	(23)
(24)			MCWB	-4.5	-3.3	0.7	10.3	14.5	18.5	18.8	18.3	13.9	8.6	2.8	-2.7	(24)
(25)		10%	DB	-6.1	-4.8	0.9	13.8	22.4	26.3	26.3	25.3	19.4	11.4	2.3	-3.9	(25)
(26)			MCWB	-6.7	-5.5	-0.2	8.8	13.7	17.3	18.4	17.5	12.8	7.6	1.0	-4.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	0.1	4.8	14.1	18.2	21.6	21.7	21.1	17.8	12.8	6.7	0.1	(27)
(28)			MCDWB	0.0	1.0	8.0	20.9	25.9	29.1	28.2	27.8	26.2	19.0	8.0	0.8	(28)
(29)		2%	WB	-2.8	-1.5	1.7	12.2	16.7	20.3	20.8	20.1	15.7	10.8	4.7	-1.1	(29)
(30)			MCDWB	-2.1	-0.7	2.8	17.6	24.4	27.4	27.3	26.7	22.3	15.3	5.9	-0.1	(30)
(31)		5%	WB	-4.5	-3.3	0.8	10.7	15.4	19.1	20.0	19.2	14.6	9.1	2.9	-2.7	(31)
(32)			MCDWB	-3.9	-2.6	1.9	16.2	22.9	26.6	26.1	25.1	20.7	13.2	4.0	-2.0	(32)
(33)		10%	WB	-6.7	-5.5	-0.2	8.9	14.2	18.1	19.2	18.1	13.3	7.7	1.1	-4.6	(33)
(34)			MCDWB	-6.2	-4.7	0.9	13.8	21.3	24.7	24.7	23.7	18.6	11.1	2.0	-3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	8.6	8.6	9.5	12.0	11.3	10.2	10.7	10.5	8.2	6.6	7.5	(35)
(36)			MCDBR	7.4	7.2	7.1	14.3	16.0	15.0	13.6	14.6	15.1	13.7	6.7	6.7	(36)
(37)			MCWBR	7.3	6.9	6.0	7.8	7.1	6.3	5.1	5.9	6.9	8.0	5.5	6.5	(37)
(38)		5% WB	MCDWR	7.5	7.2	6.7	12.8	14.1	12.6	11.3	12.2	13.2	12.2	6.3	6.9	(38)
(39)			MCWBR	7.4	6.9	5.8	7.8	7.3	5.8	5.1	5.9	6.7	7.6	5.5	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.320	0.373	0.408	0.407	0.406	0.424	0.432	0.380	0.356	0.320	0.286	(40)	
(41)		taud	2.057	2.017	1.961	2.131	2.226	2.306	2.247	2.196	2.298	2.246	2.136	2.035	(41)	
(42)		Ebn at Noon	626	734	775	811	838	843	816	781	778	697	579	527	(42)	
(43)		Edh at Noon	70	104	139	135	130	122	127	126	100	83	65	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.05	2.15	3.70	4.62	5.64	6.07	5.65	4.61	3.25	1.67	0.83	0.68	(44)	
(45)		RadStd	0.07	0.10	0.27	0.49	0.46	0.52	0.48	0.45	0.23	0.26	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page